

CONTACT

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SUMMARY

PhD researcher (Marie Curie Fellow) working on cognitively grounded and data-efficient language modeling. Specialized in LLM **mechanistic interpretability**, **retrieval-augmented generation**, and **conversational agents**. Published at NeurIPS, ICML, CogSci, and ICMI. Co-organizer of **BabyLM 2025**, **BabyLM-Speech 2026**.

SKILLS

Programming

Python (PyTorch, Tensorflow, Fairseq, TRL, vLLM, LogitLens)
Git, Bash, Linux, LaTeX

R, MATLAB, Java, Prolog, Javascript, HTML

Deep Learning

Reinforcement Learning(DPO, PPO, GRPO ...)
interpretability (logit attribution, circuit tracing, Sparse Autoencoder)

Retrieval-Augmented Generation (RAG)

Developmental benchmarking
Speech representation learning

JING LIU

PhD Researcher - AI + Cognitive Science

EDUCATION

Ph. D. - NLP & Cognitive Modeling École Normale Supérieure, Paris, France	2023 - ongoing
Thesis: Data-Efficient Spoken Language Modeling (Advisors: Emmanuel Dupoux, Najoung Kim, David Hawarh)	
Adv. Master - Artificial Intelligence KU Leuven, Leuven (Belgium)	2022 - 2023
Thesis: Computational Modeling of Early Word Acquisition.	
Research Master - Linguistics & Comm. Sciences (Distinction) Radboud University, Nijmegen (the Netherlands)	2020 - 2022
Thesis: Predicting Backchannels in Multimodal Child-Caregiver Conversation (Advisors: Abdellah Fourtassi, Asli Özyürek)	
BA in English education & BSc in Psychology East China Normal University, Shanghai, China	2015 - 2019
Thesis: Mnemonic Effects in Idiom Learning. (Advisor: Yanning Yang)	

WORK

applied scientist intern Amazon, Berlin, Germany	2026.7 - 2026.12
Anomaly detection	
research intern Inria, Paris, France	2022.2 - 2022.8
Anomaly detection	
research intern Amazon, Paris, France	2021 - 2022
Anomaly detection	
applied scientist intern Amazon, Paris, France	2026 - ongoing
Anomaly detection	

SELECTED PUBLICATIONS

Toolkits & Software

Weight & Biases, Praat, Forced Aligner, ELAN

ACHIEVEMENTS

Research Fellowships

Marie-Curie PhD Fellow

Radboud Honors Academy

National Research Training Program

Scholarships & Awards

NeurIPS Travel Fund

INTERSPEECH Student Grant

Radboud Scholarship

2nd Prize – National Teaching Competition

ECNU Guanghua Scholarship

ECNU Scholarship

- **Liu, J.**, Schweitzer, S., & Fourtassi, A. (2026). Which Forms of Caregiver Feedback Support Grammar Learning? A Reinforcement-Learning Study of Child-Like Language Models. Arxiv.
- **Liu, J.**, & Xie, S. (2026). Is Curriculum Learning Based on Pedagogical Child-Directed Speech Effective for Language Model Training? Arxiv.
- **Liu, J.**, & Shukla, A. (2026). Compression at a Cost: Interpreting Information Bottlenecks in Safety-Aligned Reward Models, Mechanistic Interpretability Workshop, The International Conference on Machine Learning (ICML), Seoul, South Korea.
- **Liu, J.**, & Kim, N. (2026). Does Episodic Memory Help Close the Lexical Frequency Gap in Sensitivity to Syntactic Contrasts? A Test Using Retrieval-Augmented Language Model, Conference On Language Modeling (COLM), San Francisco, USA.
- **Liu, J.**, & Dupoux, E. (2026). On comparing developmental curves in children and AI simulations, Annual Review of Developmental Psychology (under review).
- Charpentier, L., Choshen, L., Cotterell, R., Gul, M. O., Hu, M., Jumelet, J., Linzen, T., **Liu, J.**, Mueller, A., Ross, C., Shah, R. S., Warstadt, A., Wilcox, E., & Williams, A. (2025). Findings of the Third BabyLM Challenge: Accelerating Language Modeling Research with Cognitively Plausible Data, Proceedings of the Third BabyLM Workshop, Association for Computational Linguistics
- **Liu, J.** (2025). Layer Specialization Underlying Compositional Reasoning in Transformers, Arxiv.
- **Liu, J.** M, Lavechin., N, Hamilakis., & E. Dupoux. (2025). CORPUS-CDI: comparing children and AI models learning rate of expressive vocabulary. Arxiv.
- **Liu, J.** (2025). Circuit-Aware Reward Training: A Mechanistic Framework for Longtail Robustness in RLHF. Arxiv.
- **Liu, J.** Li, Y., & Wang, H. (2025). Distributed Specialization: Rare-Token Neurons in Large Language Models. Mechanistic Interpretability Workshop, The Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS), San Diego, the United States.
- **Liu, J.** (2025). No Clustering, No Routing: How Transformers Actually Process Rare Tokens. UniReps: 3rd Edition of the Workshop on Unifying Representations in Neural Models, The Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS), San Diego, the United States.
- **Liu, J.**, Wang, H., & Li, Y. (2025). Emergent Specialization: Rare Token Neurons in Language Models. High-dimensional Learning Dynamics Workshop, The International Conference on Machine Learning (ICML), Vancouver, Canada.
- Charpentier, L., Choshen, L., Cotterell, R., Gul, M. O., Hu, M., Jumelet, J., Linzen, T., **Liu, J.**, Mueller, A., Ross, C., Shah, R. S., Warstadt, A., Wilcox, E., & Williams, A. (2025). arXiv preprint, BabyLM Turns 3: Call for papers for the 2025 BabyLM workshop.
- **Liu, J.**, & Fourtassi, A. (2025) Benchmarking LLMs for Mimicking Child-Caregiver Language in Interaction., Cognitive Science Society, San Francisco, the United States.
- **Liu, J.**, & Lavechin, M., & Hamilakis, N. & Dupoux, E. (submitted). Machine CDI: a lexical benchmark for expressive vocabulary
- Agrawal, A., **Liu, J.**, Bodur, K., Favre, B., & Fourtassi, A. (2023). Development of Multimodal Turn Coordination in Conversations: Evidence for Adult-like behavior in Middle Childhood, Cognitive Science Society, Sydney, Australia.

- **Liu, J**, Nikolaus, M., Bodur, K., & Fourtassi, A. (2022). Predicting Backchannel Signaling in Multimodal Child-Caregiver Conversation, WoCBU Workshop, ICMI 2022(24th ACM International Conference on Multimodal Interaction), Bangalore, India

SELECTED PROJECTS

Multi-communicative feedback in child-caregiver interactions Ongoing

2026

Implemented reward aggregation and automatic weighting to align LM generations with caregiver feedback.

Evaluating speech language model in lexical discrimination tasks Ongoing

2026

Self supervised speech representation learning

MI-Reward: Mitigating Reward Hacking via Sparse Autoencoders Ongoing

2025-26

Identify and suppress spurious features in reward modeling using Sparse Autoencoders

Grokking transformer for in-context learning Ongoing

2025

Analyzed compositional reasoning with circuit tracing and quantum-inspired parameter efficient attention mechanisms.

Emergent Specialization in LLMs NeurIPS MechInterp Workshop, 2025; ICML HiLD Workshop, 2025

2025

Analyzed rare token neuron specialization using logit attribution and sparse autoencoders.

Retrieval-Augmented LMs for rare token acquisition

2024-25

Cognitive Science 2026

Investigated factors contributing to RAG system retrieval.

Communicative feedback in child-caregiver interactions Baseline model of BabyLM Challenge 2025

2024

Implemented PPO-based finetuning to align LM generations with caregiver feedback. Built baseline for BabyLM Challenge 2025.

Benchmarking LLMs for Mimicking Child-Caregiver Language in Interaction Cognitive Science 2025; metrics of BabyLM Challenge 2025

2024

Built LLM-based conversational agents to simulate child-adult interactions. Evaluated generated interactions with multi-level linguistic properties

Machine CDI Benchmark Under Review

2023-2024

Built a lexical benchmark evaluating expressive vocabulary generation in LLMs vs. children. Grounded evaluation in developmental norms.

Automatic predicting conversational behaviors ICMI 2022, Cognitive Science 2023

2022-2023

Built multi-modal model to predict conversational behaviors like backchanneling and turn-taking

WORK & SERVICES

- Area Chair, Association for Computational Linguistics, EMNLP, 2025
- Reviewer, Mech Interp Workshop at ICML, 2026
- Reviewer, BabyLM Challenge Workshop at ACL, 2026
- Co-organizer, BabyLM Challenge (speech), 2026
- Co-organizer, Computational Developmental Linguistics, ACL, 2026
- Reviewer, Association for Computational Linguistics, ACL, 2025
- Reviewer, Open Mind: Discoveries in Cognitive Science, journal
- Reviewer, Mech Interp Workshop at NeurIPS, 2025
- Reviewer, UniRep Workshop at NeurIPS, 2025
- Reviewer, BabyLM Challenge Workshop at EMNLP, 2025
- Co-organizer, BabyLM Challenge, 2025
- Invited Lecturer, Boston University, Spring 2024
- Organizer, Weekly NLP Reading Group, 2024-Now
- Teacher, Comprehensive English; Introduction to psycholinguistics, 2019-2020
- Teaching Assistant, English Phonetics, 2017-2018